PyGameMaker — Tutorial 6: Maze

How to Use This Handout

This handout goes with the in-app tutorial **Help > Tutorials > Maze: Navigate to the Exit** (5 pages, 4 core + 1 optional bonus, about 20-25 minutes for the core game). Keep the tutorial open on one half of your screen. Tick each box when the step is done, and check the **You should see** line at the end of every phase. If something does not match, look at **Stuck?** before you call your teacher.

What You Will Make

A maze game: you steer a character through corridors with the arrow keys, collect coins for points, and reach the exit to win.

Phase 1: Player and Maze

- ☐ **1.** Create two sprites (32×32): spr_player and spr_wall.
- ☐ **2.** Create obj_wall (sprite spr_wall, **Solid** ticked) and obj_player (sprite spr_player, not solid).
- ☐ **3.** In obj_player add five events: **Keyboard: Right Arrow (held)** with *Set horizontal speed to 4*; **Left Arrow (held)** with *-4*; **Down Arrow (held)** with *Set vertical speed to 4*; **Up Arrow (held)** with *-4*; and **Keyboard: No key** with *Stop movement*.
- ☐ **4.** Create four more sprites, spr_player_up/_down/_left/_right, and add a *Set sprite* block to each of the four arrow-key events from step 3 so the player's picture matches the direction it is moving.
- ☐ **5.** Add **Collision with obj_wall** with *Stop movement*.
- ☐ **6.** Create room_maze (640×480) and turn on **Snap to Grid** 32×32. Place walls around the border and inside to make corridors, and place the player at the top left. Every corridor must be at least one tile wide.

> **You should see:** press **F5**. The arrow keys move the player, its picture faces the direction you are moving, it stops at walls, and it stops when you let go.

Phase 2: Coins and Exit

- ☐ **7.** Create sprites spr_coin and spr_exit (32×32).
- ☐ **8.** Create obj_coin and obj_exit, each with its sprite and **not** solid.
- ☐ **9.** In obj_coin, add **Collision with obj_player**: *Add to score 10*, then *Destroy this instance*.
- ☐ **10.** In obj_exit, add **Collision with obj_player**: *Show message* "You Win!", then *Restart room*.
- ☐ **11.** In the room, scatter coins along the corridors and put the exit as far from the start as you can.

> **You should see:** press **F5**. Coins vanish when you touch them. Reaching the exit shows "You Win!" and the room starts again.

Phase 3: Score Display

- [ ] **12.** Create obj_game_controller (no sprite, not solid).
- [ ] **13. Create** event: *Set score to 0*. **Draw** event: *Draw score at x 10, y 10*.
- [ ] **14.** Place obj_game_controller anywhere in the room.

> **You should see:** "Score: 0" at the top left, going up by 10 for every coin. After "You Win!" the coins come back and the score is 0 again.

Phase 4: Make It 2.5D (optional bonus, page 5)

Your game is finished after Phase 3 — this is only for students who want to try something extra. No new maze needed.

- [ ] **B1.** In obj_player, add a **Create** event with *Enable Raycast View* (Cell Size 32).
- [ ] **B2.** Add one *Set Facing Angle* block to each of the same four arrow-key events from step 3/4: Right = 0, Left = 180, Up = 90, Down = 270.

> **You should see:** press **F5**. The maze now looks first-person instead of top-down, textured with your own spr_wall picture, and turning follows whichever arrow you hold — exactly like your player's picture did in Phase 1.

Stuck?

What is wrong	Most likely cause	What to do
The player does not move	The events are on the wrong object, or the player is not in the room	Check the five events on obj_player; check the room
The player slides forever	The **No key** event is missing	Add **Keyboard: No key** with *Stop movement*
The player's picture never changes	A **Set sprite** block is missing from one of the four arrow-key events	Add it, matching the direction
The player walks through walls	obj_wall is not Solid, or the player has no **Collision with obj_wall**	Tick **Solid**; add the event with *Stop movement*
The player gets stuck in corridors	Walls or the player were not placed on the grid	Turn on **Snap to Grid** 32×32 and place them again
Coins do nothing	The coin has no **Collision with obj_player** event	Add it to obj_coin
The whole game freezes at the exit	*Show message* is waiting for a click	Click OK on the message
No score on the screen	obj_game_controller is not in the room	Place it in the room
The score does not go back to 0 after winning	*Set score to 0* is missing in the controller's **Create**	Add it

(Bonus) The maze still looks top-down	*Enable Raycast View* is missing, or the 2.5D Raycast View extension is not checked in **File > Project Settings... > Extensions**	Add the Create event; check the extension
(Bonus) The camera does not turn	A *Set Facing Angle* block is missing from one of the four arrow-key events	Add it, matching the table in step B2

Challenges

- **Try this (5 minutes):** change the player's speed, or the value of a coin.
- **Push further:** make a second maze in another room and use **Go to next room** at the exit.
- **Invent:** add a patrolling enemy, a countdown timer, or a key that must be collected before the exit works.

Vocabulary

Term	What it means
Solid	An object that blocks movement
Collectible	An object that disappears when the player touches it
Exit	An object that ends or advances the level
Snap to Grid	Placing objects exactly on a regular grid
Level design	Planning the maze so it is fair and solvable
Facing direction	Which way a character is pointing; used to pick a picture (Phase 1) or a camera angle (bonus)

Check Yourself

Write your answers in the notes below.

1. Why does the player need a **No key** event with *Stop movement*?
2. Why does the coin destroy *this* instance, and what would *Destroy other instance* have done in the coin's event?
3. Why do we place objects on a grid when we build the maze?

My Notes